Ribosomal RNA dominance and unannotated mitochondrial rDNA confound 3′-tag RNA-seq in *Pleurotus ostreatus*: a corrected reference, annotation and metabolic reconstruction

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Abstract

Transcriptomic studies of non-model basidiomycetes increasingly rely on 3′-end tag RNA-seq and on public genome assemblies whose annotations were built for gene finding rather than for read quantification. We show that both choices carry substantial and largely silent costs. From 92.4 million reads across 16 libraries of *Pleurotus ostreatus* cv. "Harbor Blue P01" grown in a mycoponic ceramic-tube system, only 2.48 million reads (2.7%) were assignable to protein-coding features. Nuclear ribosomal DNA accounted for a mean of 53.5% of reads, and the single largest remaining component, mitochondrial rRNA, is unannotated in all four public *P. ostreatus* genome annotations we examined. Because it carries no feature it is neither counted nor removable by the biotype-based rRNA filtering that current consensus pipelines implement; it is silently discarded as "no feature". We further show that the intuitive remedy for low assignment in tag data—extending gene 3′ ends—recovered 0.1% here, and we give the diagnostic that distinguishes truncated UTRs from unannotated rRNA. Assembly choice also proved consequential in an unexpected way: the RefSeq-designated reference genome for the species performed worst of four candidates (5.45% versus 24.94% uniquely mapped), and assemblies differ in whether they resolve or collapse the nuclear rDNA array, which changes multi-mapping and therefore quantification. We provide a corrected, rRNA-complete and 3′-aware annotation, a functional annotation of the proteome including 423 CAZymes and a predicted secretome, and the first draft genome-scale metabolic reconstruction for *P. ostreatus*, which achieves non-zero biomass flux on the defined culture medium. Despite the compromised input, the corrected pipeline recovered convergent biological signal in two tissues by two independent criteria, demonstrating both what such data can still support and where the limit lies.

Keywords: *Pleurotus ostreatus*, 3′-tag RNA-seq, ribosomal RNA, genome annotation, genome-scale metabolic model, WGCNA

INTERNAL DRAFT

This version retains the full account of the sequencing provider's errors (Section 5.2) and is intended for internal review and for the provider conversation. Two derived manuscripts omit that material: a resource paper and a separate biology paper on the exudophore. All numeric values are generated from the analysis repository and verified by `scripts/32_audit_numbers.py`.

1 Introduction

Pleurotus ostreatus is among the most widely cultivated edible fungi and a workhorse model for white-rot lignocellulose degradation. Its enzymatic repertoire—laccases, manganese and versatile peroxidases, lytic polysaccharide monooxygenases and an extensive glycoside hydrolase complement—underpins both its ecological role and a growing set of biotechnological applications. Interest has recently extended to controlled-environment and bioregenerative life support contexts, where fungal biomass offers a protein source that can be produced from lignocellulosic residues in a closed system. The mycoponic culture format used here, in which mycelium is grown on micro-structured ceramic tubes supplied with liquid nutrient medium through an antimicrobial size-exclusion interface, was developed for exactly this purpose [1].

Understanding how such a system works requires resolving what different parts of the mycelium are doing. Filamentous fungal colonies are not homogeneous: they differentiate into morphologically and physiologically distinct regions, and in this culture system they form several visually distinguishable tissue types, including structures that produce and exude liquid droplets.

For multi-condition designs of this kind, 3′-end tag counting has become an attractive option: it sequences a single fragment per transcript, so cost per sample falls sharply and library complexity requirements are lower than for full-length protocols. The trade-offs are less widely appreciated than the benefits. Because one read corresponds to one molecule regardless of transcript length, no length normalisation applies and TPM or FPKM values are meaningless. Because reads pile up at the polyadenylation site, quantification depends on the accuracy of annotated 3′ ends rather than on gene bodies. And because the method primes on poly(A) tracts, it is sensitive to internal priming and to any failure of poly(A) selection.

Those dependencies interact badly with a second issue: the state of public fungal genome annotation. Genome annotations are produced to describe protein-coding gene models. Ribosomal DNA is frequently omitted, being repetitive, hard to assemble and of little interest for gene finding; organellar rRNA is omitted more often still. For a genome browser this is inconsequential. For read quantification it is not, because a feature that does not exist cannot receive counts and cannot be filtered out. Current consensus RNA-seq pipelines, including the NASA GeneLab pipeline used as the basis for this work [2], implement rRNA removal by dropping features whose annotated biotype is rRNA—a strategy that is silently ineffective against rRNA that was never annotated.

We encountered both problems in an acute form, and use them to quantify costs that are usually invisible and to provide the corrected resources that the species has lacked.

2 Results

2.1 Reference selection changes mapping by an order of magnitude

Four *P. ostreatus* assemblies were compared empirically rather than assumed (Table 2, Fig. 2a). Uniquely mapped fraction differed far more than assembly statistics would suggest. The RefSeq-designated reference genome for the species (PC9) performed *worst*, at 5.45% uniquely mapped against 24.94% for the best candidate—less than a quarter. BOM_ss5 and BOM_ss14 differed by 0.02 percentage points, consistent with their being the two nuclei of a single dikaryon, matching the cultivar used here.

On matched 3′-extended annotations, BOM_ss5 assigned 6.9% more reads than PC9.15 and did so for every one of the 16 libraries, with gains from 4.0% to 13.5% (Fig. 2b). BOM_ss5 was used for all subsequent analysis; the complete parallel PC9.15 analysis is retained in the repository.

2.2 The libraries are dominated by rRNA from two distinct sources

Against barrnap-derived loci, nuclear rDNA accounted for a mean of 53.5% of reads (range 19.9–69.7%; Fig. 1b). This is a direct measurement; a k-mer probe estimate made earlier in the analysis (12–46%) was a substantial underestimate, and we note this because probe-based estimates are common in quality-control reporting.

Clustering intergenic reads from pooled alignments revealed the second source. In BOM_ss5, 59.7% of pooled aligned reads fall in a tandem block structure at `JBQVBD010000012.1:2,298,112--2,346,320` (the nuclear rDNA array) and 29.1% on the mitochondrial contig `CM148777.1`—approximately 89% of aligned reads in identifiable rDNA loci (Supplementary Fig. S7). barrnap's fragmentary 5S/5.8S/18S/28S calls fall inside every block, confirming their identity. In PC9.15 the equivalent mitochondrial block contained 60.9% of all pooled aligned reads and was confirmed by BLAST at 100% identity to the *P. ostreatus* mitochondrion.

Neither the nuclear rDNA array nor the mitochondrial rRNA is annotated as a feature in any of the four public annotations. In the GTF, the mitochondrial rRNA region is flanked only by tRNA gene models (Fig. 2c). The practical consequence is severe: these reads cannot be counted, cannot be removed by biotype-based rRNA filtering, and are silently reported as "no feature".

2.3 Assembly choice changes rDNA behaviour, not just contiguity

BOM_ss5 resolves multiple tandem copies of the nuclear rDNA repeat; PC9.15 collapses them. The consequence is quantitative, not cosmetic: in the collapsed assembly, rDNA reads distribute across fewer loci and are reported as multi-mapping and discarded, whereas in the resolved assembly they pile into an identifiable block that can be annotated and accounted for. This is the mechanism underlying the assignment-rate difference in Section 2.1, and it means that assembly selection for RNA-seq should consider rDNA representation, which is not reported in standard assembly metrics.

2.4 Extending gene 3′ ends does not fix the problem

A natural response to low assignment in 3′-tag data is that annotated 3′ UTRs are too short. This is superficially well supported here: of the four annotations, only PC9.15 carries meaningful UTRs (median 3′ UTR 78 bp; the other three have 3 bp, i.e. the stop codon only).

We tested the idea on PC9.15, which as the only annotation with meaningful UTRs is the most favourable case for it. Extending gene 3′ ends strand-aware by up to 500 bp with a neighbour cap lengthened 91.3% of its 13,556 genes (median extension 323 bp; 3.79 Mb added). The effect on gene assignment was 15.5% $\rightarrow$ 15.6% (Supplementary Fig. S2). The same procedure on BOM_ss5, where it forms part of the final annotation, extends 97.7% of 12,705 genes by a median of 500 bp and adds 4.55 Mb.

The diagnostic that distinguishes the two causes is simple and we recommend it as routine: *cluster the reads that fail to be assigned and ask whether they are diffuse or focal*. UTR truncation produces diffuse loss distributed just downstream of many genes; unannotated rRNA produces a small number of very high-coverage blocks. Here a single locus held 60.9% of aligned reads. Adding rRNA features reduced “no feature” from 30–78% to 0–1.2%.

2.5 Effective yield and the cost of retaining noise-dominated libraries

Despite 92,430,743 raw reads, only 2,475,808 were assigned to non-rRNA features across all 16 libraries; 7,601,974 were assigned to rRNA features. Effective mRNA yield ranged from 0.09% to 10.49% of raw reads, and only 3 of 16 libraries exceeded 100,000 assigned non-rRNA counts (Table 1, Fig. 1).

Retaining the lowest-yield libraries actively degraded the analysis. With all 16, the first principal component correlated with sequencing depth at $r = 0.681$—the dominant axis of variation was library size, not biology. Excluding the four libraries below 15,000 assigned non-rRNA counts reduced this to $r = -0.003$, while increasing genes passing expression filtering from 1,199 to 1,666 and genes with a significant tissue effect from 82 to 281 (Fig. 3). All four tissues retained $n \geq 2$.

2.6 Resources generated

Annotation. rRNA-complete, 3′-aware GTFs for BOM_ss5 and PC9.15, with rRNA blocks declared in per-reference tables.

Functional annotation. Of 12,521 BOM_ss5 proteins, 5,163 (41%) have a Swiss-Prot hit; 39,607 GO/KEGG term assignments; 423 proteins carrying a CAZy-class domain (GH 233, AA 86, GT 57, PL 30, CBM 13, CE 4); 576 proteins predicted secreted (Supplementary Fig. S4). The auxiliary-activity complement—laccases, peroxidases, lytic polysaccharide monooxygenases—is consistent with the oak sawdust and microcrystalline cellulose in the growth medium.

Metabolic reconstruction. The first draft genome-scale metabolic model for *P. ostreatus*: 1,915 reactions, 2,028 metabolites and 881 genes, with gene–protein–reaction rules traceable to supporting proteins. Constraining exchanges to the mycoponic medium and adding transport reactions raised the number of reactions able to carry flux from 288 to 2,262. Targeted gapfilling then completed the cofactor pathways with three added reactions carrying no gene evidence; biotin remained unreachable and was supplied by the medium rather than given a biosynthesis route, which is biologically appropriate for a likely auxotroph. The final model has 5,247 reactions of which 2,287 carry flux, all six tested cofactors producible, and non-zero biomass flux on the defined medium (Table 6, Supplementary Fig. S5).

2.7 The pipeline recovers coherent signal where power permits

We assessed tissue signal by two independent criteria.

Replicate-supported markers. Tissue specificity was scored with the τ index on linear normalised counts. Because τ is computed on tissue means, a single outlying replicate can manufacture an apparent marker; we therefore required that *every* replicate of a tissue exceed the maximum value in any sample of any other tissue. Of the top 50 markers per tissue, this was satisfied by 37 for exudophore ($n = 2$), 22 for nodule ($n = 3$), 1 for exuding mycelium ($n = 3$) and 1 for fuzzy mycelium ($n = 4$) (Supplementary Fig. S3). The test becomes more stringent as n increases, so these counts are not comparable across tissues; the qualitative separation is nonetheless clear.

Co-expression modules. WGCNA (12 libraries, 4,748 genes) produced 41 modules, 37 with at least one enriched GO or KEGG term. Of 168 module $\times$ tissue correlations, 17 reached $p < 0.05$ (8.4 expected by chance) and six survived Benjamini–Hochberg correction—all associated with exudophore or nodule (Table 3, Fig. 4a).

The two methods agree completely on which tissues carry signal, despite being sensitive to different failure modes. Neither supports exuding or fuzzy mycelium at the achieved depth.

Convergent signatures. The nodule-associated `brown` module enriches for extracellular space (7/20 genes, FDR = 0.003) and CH-OH-acting oxidoreductases, and contains 22 predicted secreted proteins and 11 CAZymes; it also enriches for “protein aggregate centre” (3/5). The replicate-supported nodule markers are independently dominated by cerato-platanin family proteins and small heat-shock proteins—consistent with the aggregate-centre enrichment recovered by a different method.

The exudophore, a newly described exudate-producing structure, yields the largest set of replicate-supported markers of any tissue. These are consistently oxidative and secretory: alcohol oxidase 1, glyoxal oxidase, formate and aldehyde dehydrogenases, an FAD-dependent monooxygenase, acetate–CoA ligase, an amino-acid permease and an AA9 lytic polysaccharide monooxygenase. Alcohol oxidase and glyoxal oxidase both generate extracellular H_2O_2, the co-substrate required by fungal peroxidases. We report these as evidence that the corrected pipeline recovers coherent, method-independent signal, not as a characterisation of exudophore function; the full tissue comparison is reserved for a companion study.

2.8 Tissue modules are not explained by ribosomal RNA load

Because rRNA content varies among these libraries (42.0–96.6% of assigned counts) and is itself correlated with usable depth ($r = -0.907$ with $\log_{10}$ mRNA counts across the twelve libraries entering the network), any module correlating with tissue could in principle reflect rRNA load rather than biology. This concern is sharpest for the exudophore-associated `lightcyan` module, which enriches for SSU-rRNA maturation—a ribosome-biogenesis signature that could plausibly co-vary with ribosomal content for purely technical reasons.

We tested each module eigengene against the per-library rRNA fraction and recomputed the tissue association as a partial correlation controlling for it (Fig. 4b, Table 3). All six associations survive. The `lightcyan` module correlates with rRNA fraction at only -0.15, so there is essentially no technical signal to remove. Because rRNA fraction is strongly anti-correlated with usable depth, this control also partially accounts for depth.

3 Discussion

3.1 Unannotated rRNA is a silent failure mode

The largest single component of these libraries occupied a genomic region carrying no annotated feature, and its consequences were entirely silent. No tool raised an error. Alignment rates looked unremarkable. The reads were reported under `Unassigned_NoFeatures`, a category most workflows summarise but few interrogate.

The consequence for rRNA removal deserves emphasis. Version G of the GeneLab consensus pipeline added a parallel rRNA-removed differential expression track, implemented by dropping features whose biotype is rRNA before renormalisation [2]. That is a sound design, and we adopted it. But it is a *feature*-based operation, and it therefore cannot remove rRNA that was never annotated as a feature. A pipeline can execute its rRNA-removal step faithfully, report success, and leave the majority of the ribosomal signal untouched. The vulnerability grows precisely where it matters most—a non-model organism whose annotation was contributed by a genome project rather than curated for expression analysis.

Two properties made this detectable: the loss was focal rather than diffuse, and the identity was verifiable. We suggest that clustering unassigned reads and inspecting the top loci should be routine when assignment rates are low; it costs minutes.

3.2 The intuitive explanation was wrong, and testing it was cheap

Low gene assignment in 3′-tag data has an obvious candidate explanation, and it was well supported here. Acting on it would have been reasonable. Testing it took an afternoon and showed it to be almost entirely wrong (Section 2.4). Had we adopted the extension without measuring its effect, we would have concluded that the annotation was now adequate, retained a plausible-sounding correction with no benefit, and never looked for the real cause. The generalisable point is not that UTR extension is useless—it remains correct practice for tag data, and we retain it—but that the acceptance criterion should be specified before the correction is applied.

3.3 Reference choice is consequential, and standard metrics do not predict it

The RefSeq-designated reference genome produced the worst mapping of four assemblies, by a factor of more than four, and contiguity did not predict performance either: a contig-level assembly outperformed a chromosome-level one on gene assignment. The property that mattered is not reported by any standard assembly metric. For expression work on species with several published assemblies we recommend empirical selection on assignment rate using matched annotations, rather than selection on assembly designation, contiguity or gene count.

3.4 What poly(A) selection did not do

Poly(A) selection is expected to deplete ribosomal RNA, which is not polyadenylated. Here it did not, and the residual ribosomal fraction varied systematically with tissue. Fungal tissue with high ribosome content presents a harder depletion problem than the cell types for which such protocols are typically validated; where sample types of this kind are being profiled, explicit ribosomal depletion may be the safer choice, and a pilot library is a cheap way to establish which is needed. Because ribosomal content covaried with tissue identity, it was confounded with the biological factor of interest, which makes the rRNA-removed analysis track a requirement rather than a refinement—and that track could only be implemented correctly once the missing rRNA features had been added.

3.5 Homology-based annotation misses the interesting proteins

Forty-one per cent of the proteome carried a Swiss-Prot match. That figure sets a hard ceiling on every downstream enrichment analysis and explains why the two largest co-expression modules had no enriched terms at all: they are dominated by proteins with no characterised homologue. Lineage-specific secreted proteins are precisely the class most likely to be involved in a novel exudate-producing structure, and precisely the class least likely to have a Swiss-Prot homologue. Our secretome estimate is a lower bound for the same reason, and absence of enrichment in these data is weak evidence.

3.6 What compromised data can still support

Roughly 2.7% of sequenced reads reached a protein-coding feature. Establishing where the limit falls required explicit tests rather than judgement. Including the four lowest-yield libraries made sequencing depth the dominant axis of variation; removing them eliminated the artefact entirely and *increased* the number of genes passing expression filtering and the number with a detectable tissue effect. Noise-dominated libraries do not merely add nothing; they distort dispersion estimation for every other sample. Retaining all samples is not the conservative choice.

We then applied two criteria sensitive to different failure modes—a marker test guarding against single-replicate artefacts, and FDR-corrected module–trait association guarding against multiplicity. They agreed completely. Where such criteria disagree, neither should be trusted; where they converge, the conclusion is considerably stronger than either alone. Both tissues that failed had four and three replicates respectively, more than one that passed: sequencing depth, not replicate number, determined what was recoverable.

3.7 The metabolic reconstruction

The draft model is offered as a community starting point rather than a predictive tool. Its construction exposed two limitations worth recording. Mapping enzyme commission numbers onto a reaction database recovers core metabolism well but systematically misses steps annotated without a complete EC, which fell disproportionately on cofactor biosynthesis; and transport reactions, which carry no EC at all, were absent entirely from the initial draft and accounted for most of its blocked reactions. One gapfilling case is instructive: biotin remained unreachable, and we added it to the medium rather than to the network, because many fungi are biotin auxotrophs and the culture medium supplies B-vitamins. Automated gapfilling will readily invent a biosynthetic route for any metabolite declared essential; whether it should is a biological question, not an algorithmic one.

3.8 Limitations

Differential expression rests on 12 libraries with two to four replicates per tissue, and only three libraries exceeded 100,000 assigned counts. The co-expression network was constructed at a sample size below that recommended by the method's authors [3], and its modules are exploratory. CAZy assignments are Pfam-derived class-level calls, not dbCAN family assignments. The secretome is homology-transferred, not predicted *de novo*. Conclusions regarding two of the four tissues are absent not because those tissues lack distinct biology but because these libraries could not resolve it.

4 Conclusions

Ribosomal RNA absent from a genome annotation is invisible to the rRNA-removal steps that current consensus pipelines implement, and its loss is reported in a category that is rarely examined. Reference selection for expression analysis should be empirical; the species reference genome performed worst of four candidates here, and the property that determined performance is not reported by standard assembly metrics. Severely compromised data can still support conclusions, but only those that survive criteria sensitive to different failure modes, and only after noise-dominated libraries are removed rather than retained out of caution.

5 Methods

5.1 Biological material and culture

Pleurotus ostreatus cv. "Harbor Blue P01" was grown on micro-structured ceramic MiniTubes (10 cm × 5 cm, pore size <300 nm, 50% v/v granular activated carbon) supplied with mycoponic nutrient medium v3, at 16 °C, 85% relative humidity and CO_2 ≤1000 ppm [1]. Four tissue types were sampled with four biological replicates each: exuding mycelium, fuzzy mycelium, exudophore (a newly described exudate-producing structure), and nodule.

[AUTHORS: sampling procedure, tissue definitions, morphological description of the exudophore, RNA extraction method, and culture age at sampling.]

5.2 Sequencing, and its quality

Libraries were prepared and sequenced commercially as poly(A)-selected 3′-end tag libraries (Illumina NovaSeq X Plus, single-end 94 bp, 14 nt unique molecular identifier appended to the read name). All 16 libraries were sequenced on one flowcell and lane, so no batch term was required. Total yield was 92,430,743 raw reads (median 5.45 M per library).

We report the following quality observations in full, because they materially constrain the conclusions and are not evident from the delivered quality-control outputs.

The delivered analysis was aligned to the wrong species. The provider's expression matrix contains *Saccharomyces cerevisiae* systematic gene identifiers; its 7,127 features and 6,600 protein-coding genes correspond exactly to the SGD R64 annotation. Uniquely mapped reads numbered 453–2,284 per library, or 0.006–0.03% of input. The accompanying gene-set enrichment analysis used a human collection. The delivered expression matrix, principal component analysis, correlation heatmap, biotype summary and differential expression results are consequently without meaning. The reads themselves are sound: the most abundant read matches *Pleurotus* 28S rRNA at 100% identity.

Sequencing depth fell short of specification. All 16 libraries were delivered at 11–62% of the 20 M raw reads per sample advertised for the service (median 27%).

Poly(A) selection did not enrich mRNA. Ribosomal RNA constituted 20–70% of reads per library. This is a property of the delivered libraries rather than of the downstream analysis.

These observations are separable in kind. The species mis-assignment is an analytical error in the delivered report and does not affect the underlying reads. The depth shortfall is a quantitative deviation from specification. The ribosomal content is a library-preparation outcome that neither party detected before delivery, and which the provider's own summary statistics—reporting approximately 16% of reads as "mapped" without comment—would have flagged had the reference been correct. The general lesson is that provider quality-control metrics computed against a wrong reference are not merely uninformative but actively misleading.

5.3 Reference selection

Four assemblies were compared (Table 2); 250,000 reads per library were mapped to genome-only indices and compared on uniquely mapped fraction, then on gene assignment rate using matched 3′-extended annotations.

5.4 Annotation construction

Ribosomal RNA loci were identified with barrnap [4] in nuclear and mitochondrial modes and supplemented with coverage-derived blocks obtained by clustering intergenic reads from pooled alignments. Gene 3′ ends were extended strand-aware by up to 500 bp, capped at the distance to the nearest neighbouring feature minus 50 bp. rRNA blocks were added as `gene_biotype "rRNA"` features on a single strand; annotating both strands makes every overlapping read ambiguous under unstranded counting.

5.5 Read processing and quantification

Adapted from GL-DPPD-7101-G [2]. Reads were trimmed with fastp 1.3.6 [5] (polyG, polyX, 3′ quality, minimum length 30), aligned with HISAT2 2.2.3 [6] (`--max-intronlen 3000 --rna-strandness F`), reduced to primary alignments with SAMtools 1.24 [7] (`-F 0x104`), deduplicated with UMI-tools 1.1.6 [8] (`--method=directional`), and counted with featureCounts (Subread 2.1.1) [9] (`-t exon -g gene_id -s 1`). Strandedness was established empirically (forward: 49.6% assigned versus 6.3% reverse).

Two deviations from the source pipeline are consequential. HISAT2 replaces STAR because STAR was non-functional on the analysis platform, reproducibly reporting zero input reads including on a synthetic control. featureCounts replaces RSEM because RSEM models full-length transcript coverage and is invalid for 3′-tag pileups. No TPM or FPKM values are reported anywhere.

5.6 Differential expression

DESeq2 1.50.2 [10] with `~ Tissue`, running two parallel tracks (all-genes and rRNA-removed, each independently renormalised) as GL-DPPD-7101-G specifies. Genes were filtered with `edgeR::filterByExpr` [11]. Libraries yielding fewer than 15,000 assigned non-rRNA counts were excluded in a sensitivity analysis.

5.7 Functional annotation

DIAMOND [12] blastp against UniProt Swiss-Prot [13]; EC, GO and KEGG [14] assignments derived from matched Swiss-Prot records. CAZy classes were assigned from Pfam-A [15] domains detected with HMMER [16] using curated keyword rules; these are class-level, Pfam-derived calls, not dbCAN family assignments. Secreted proteins were inferred by transferring curated SIGNAL and TRANSMEM features from the best-hit Swiss-Prot entry.

5.8 Co-expression network

WGCNA 1.74 [3], signed network, soft-threshold power 18, minimum module size 30, merge height 0.25 (Supplementary Fig. S1). Module–trait correlations were corrected across all tests with Benjamini–Hochberg [17]. Enrichment used the hypergeometric test with BH correction within each module $\times$ ontology.

5.9 Metabolic reconstruction

EC numbers were mapped onto the ModelSEED biochemistry database [18] to produce a draft with gene–protein–reaction rules traceable to supporting proteins. Transport reactions were added and exchange bounds constrained to the culture medium. Targeted gapfilling restricted the candidate pool by backward reachability from each blocked metabolite. Analyses used COBRApy 0.32.1 [19].

Figures

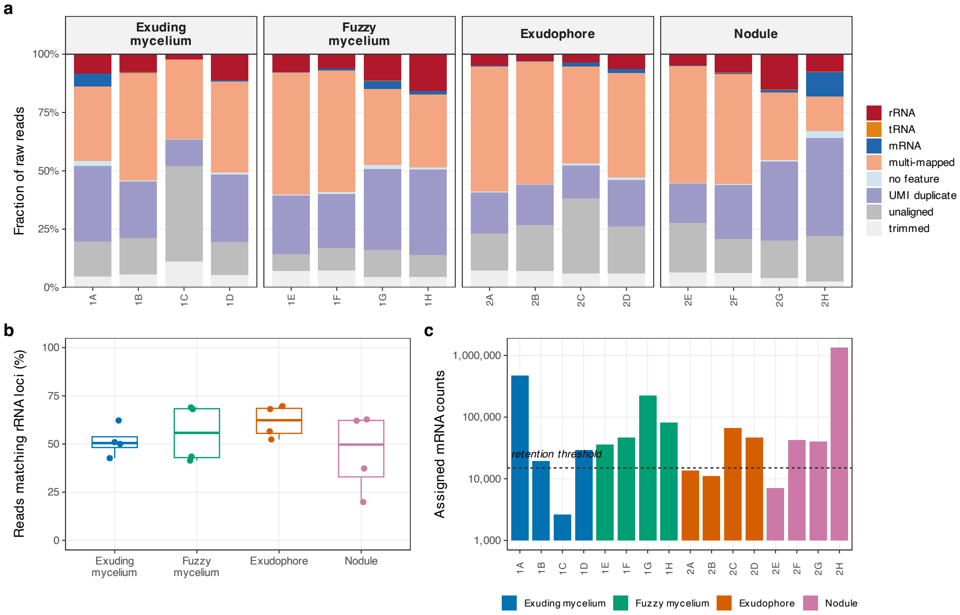

Fig. 1 Read fate and ribosomal composition. a, Fate of every raw read, reconciled exactly per library. **b**, Directly measured rRNA content, from subsampled reads mapped against the rRNA loci themselves, independently of the counting annotation. **c**, Assigned mRNA counts per library; dashed line marks the 15,000-count retention threshold.

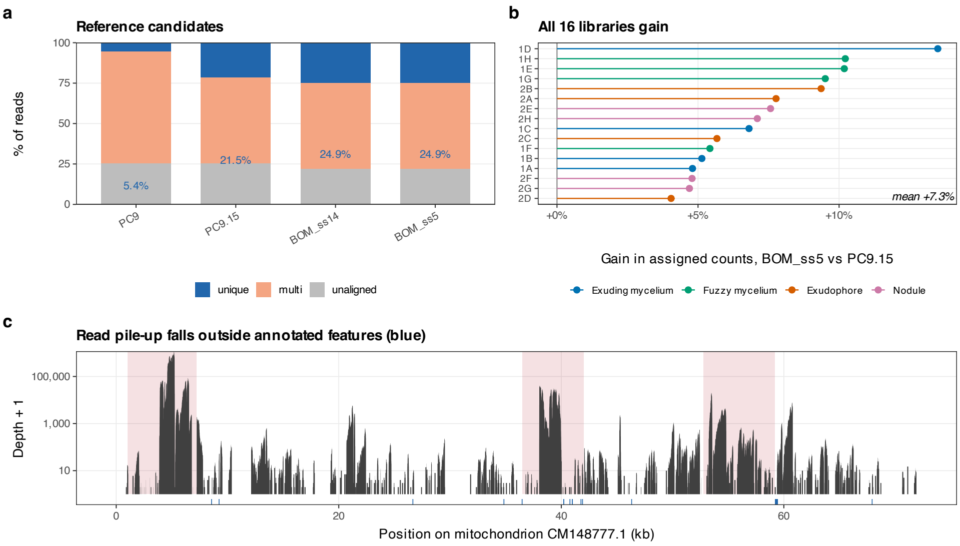

Fig. 2 Reference selection and unannotated ribosomal DNA. a, Mapping outcome for four candidate assemblies; the RefSeq reference genome (PC9) performs worst. **b**, Per-library gain in assigned counts from BOM_ss5 relative to PC9.15; all 16 libraries gain. **c**, Read depth across the mitochondrion. Shaded blocks are the coverage-derived rRNA regions added here; blue marks below the axis are features annotated in the public GTF. The highest-coverage regions carry no annotation.

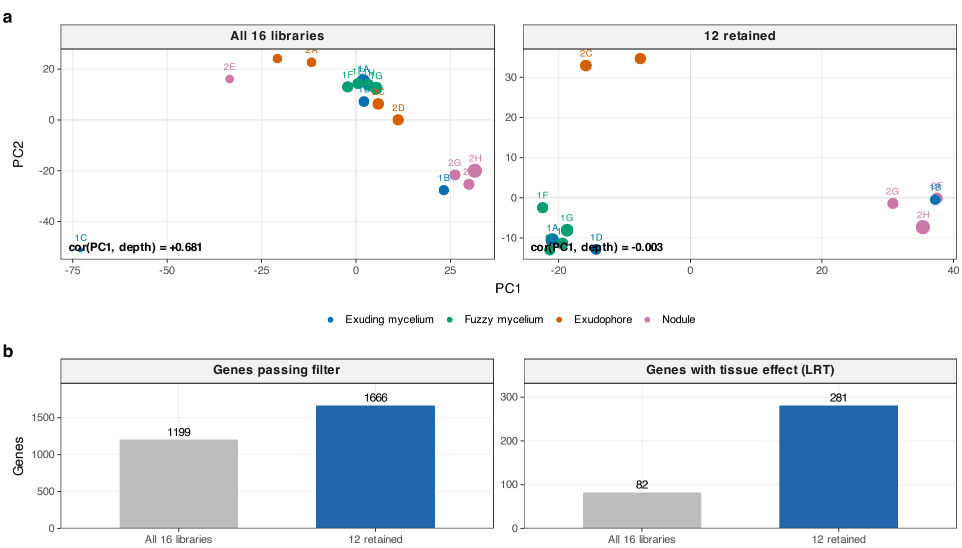

Fig. 3 Excluding noise-dominated libraries removes a depth artefact. a, VST principal components before and after excluding four libraries below 15,000 assigned non-rRNA counts; point size is $\log_{10}$ depth. **b**, Genes passing expression filtering and genes with a significant tissue effect both increase.

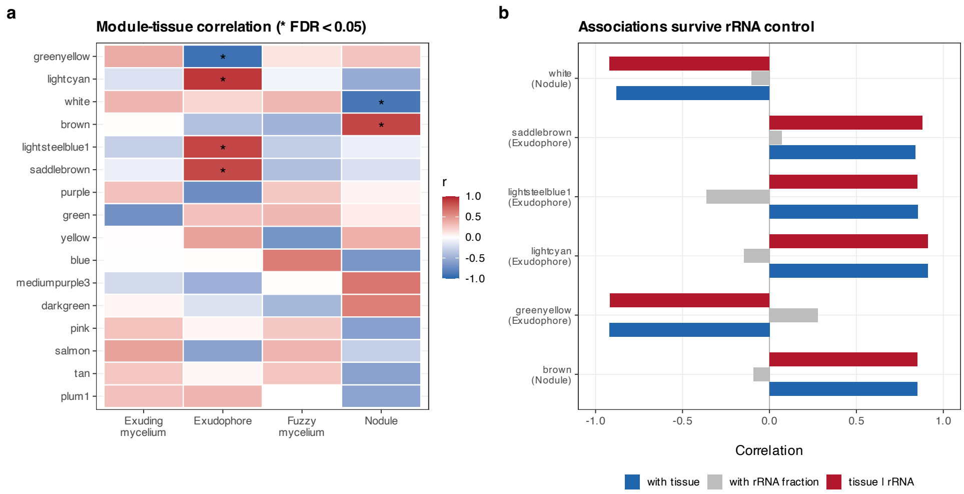

Fig. 4 **Co-expression modules.** **a**, Module–tissue correlation; asterisks mark associations surviving Benjamini–Hochberg correction across all 168 tests. **b**, Each surviving association compared with its correlation to per-library rRNA fraction, and recomputed as a partial correlation controlling for it.

Tables

Table 1 Per-library read budget against the BOM_ss5 reference after UMI deduplication. rRNA and mRNA are counts assigned to features of the respective biotype.

Tissue	Well	Raw reads	rRNA	mRNA	mRNA %	Genes ≥10	
Exuding mycelium	1A	8,423,109	701,579	469,981	5.58	6,017	
Exuding mycelium	1B	5,902,765	459,570	19,365	0.33	354	
Exuding mycelium	1C	2,770,639	64,744	2,620	0.09	19	†
Exuding mycelium	1D	5,770,952	646,755	29,276	0.51	546	
Fuzzy mycelium	1E	13,369,308	1,032,480	35,940	0.27	704	
Fuzzy mycelium	1F	5,289,692	330,615	46,085	0.87	1,034	
Fuzzy mycelium	1G	6,274,968	714,848	224,213	3.57	4,507	
Fuzzy mycelium	1H	5,610,507	887,407	81,814	1.46	1,698	
Exudophore	2A	2,941,102	145,479	13,447	0.46	239	†
Exudophore	2B	3,412,844	99,368	10,944	0.32	164	†
Exudophore	2C	4,028,384	143,150	66,739	1.66	1,485	
Exudophore	2D	2,620,662	168,385	46,606	1.78	998	
Nodule	2E	2,431,986	118,075	7,062	0.29	109	†
Nodule	2F	7,338,750	588,323	41,959	0.57	827	
Nodule	2G	3,473,216	531,658	40,253	1.16	813	
Nodule	2H	12,771,859	969,538	1,339,504	10.49	7,512	

† excluded from the differential expression analysis (fewer than 15,000 assigned non-rRNA counts).

Table 2 Reference candidates compared empirically. 250,000 reads per library mapped to genome-only indices; values are means across all 16 libraries.

Assembly	Accession	Level	Genes	Unique %	Multi %	Unaligned %
BOM_ss5	GCA_056149245.1	contig	12,705	24.94	53.17	21.89
BOM_ss14	GCA_056149315.1	contig	13,310	24.92	53.18	21.90
PC9.15	GCA_029852705.2	chromosome	13,556	21.49	53.10	25.42
PC9	GCF_014466165.1	contig	11,849	5.45	69.14	25.41

The RefSeq-designated reference genome for the species (PC9) performed worst.

Table 3 Co-expression modules whose tissue association survives Benjamini–Hochberg correction across all 168 module $\times$ tissue tests.

Module	Genes	Tissue	r	FDR	r_{rRNA}	r_{partial}	Enriched terms
greenyellow	142	Exudophore	-0.92	0.0034	+0.28	-0.92	pexophagy / response to calcium ion
lightcyan	99	Exudophore	+0.91	0.0034	-0.15	+0.91	maturation of SSU-rRNA / CENP-A containing chromatin...
white	55	Nodule	-0.88	0.0091	-0.10	-0.92	one-carbon metabolic process
lightsteelblue1	34	Exudophore	+0.85	0.0155	-0.36	+0.85	chromosome segregation
brown	320	Nodule	+0.85	0.0155	-0.09	+0.85	oxidoreductase activity / oxidoreductase activity, a...
saddlebrown	49	Exudophore	+0.84	0.0179	+0.07	+0.88	Methane metabolism / Microbial metabolism in diverse...

r_{rRNA}, correlation of the module eigengene with per-library rRNA fraction; r_{partial}, tissue correlation controlling for it.

Supplementary Figures

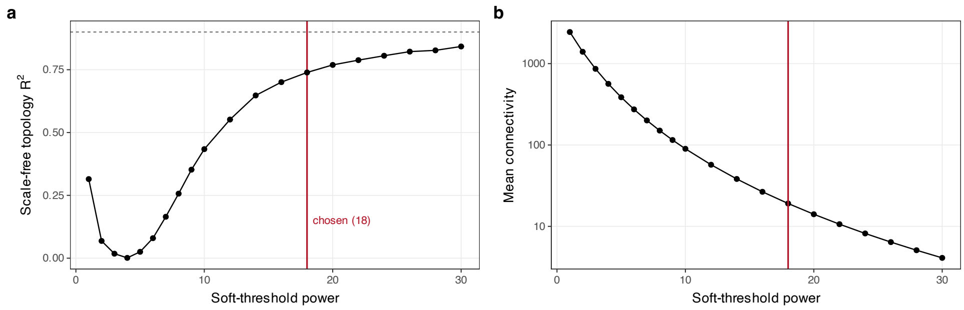

Fig. S1 Co-expression network construction diagnostics. **a**, Scale-free topology fit against soft-threshold power; no power reaches the conventional $R^2 = 0.9$ at this sample size, and 18 was used. **b**, Mean connectivity.

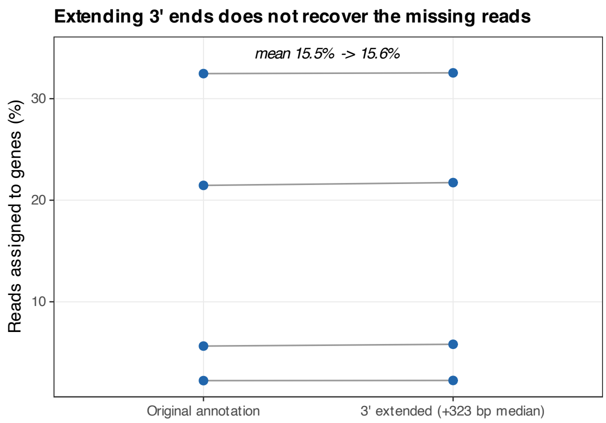

Fig. S2 Extending gene 3′ ends does not recover the missing reads. Per-library assignment rate before and after strand-aware 3′ extension of 91.3% of PC9.15 genes by a median of 323 bp.

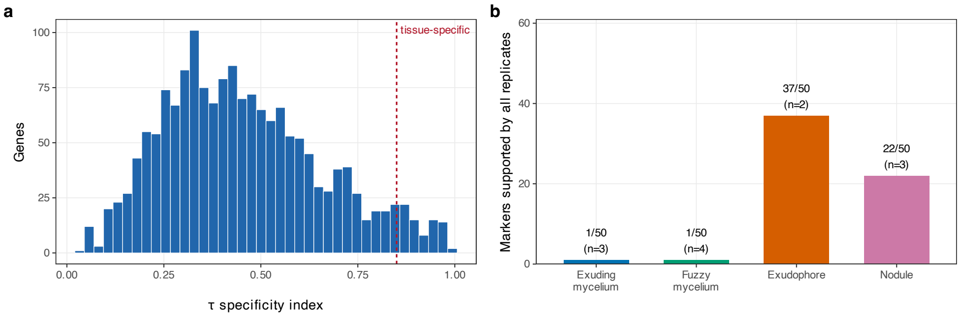

Fig. S3 Specificity scores and replicate support. **a**, Distribution of the τ index. **b**, Of the top 50 markers per tissue, the number supported by every replicate.

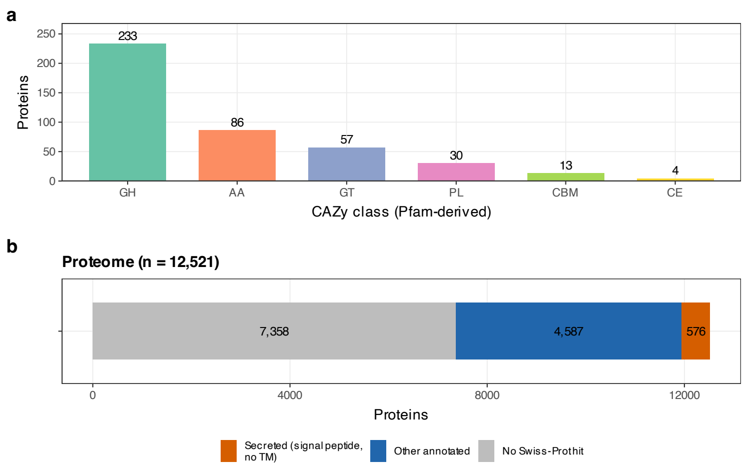

Fig. S4 Carbohydrate-active enzymes and predicted secretome. **a**, CAZy classes assigned from Pfam domains. **b**, Proportion of the proteome with a Swiss-Prot homologue and, of those, the predicted secreted fraction.

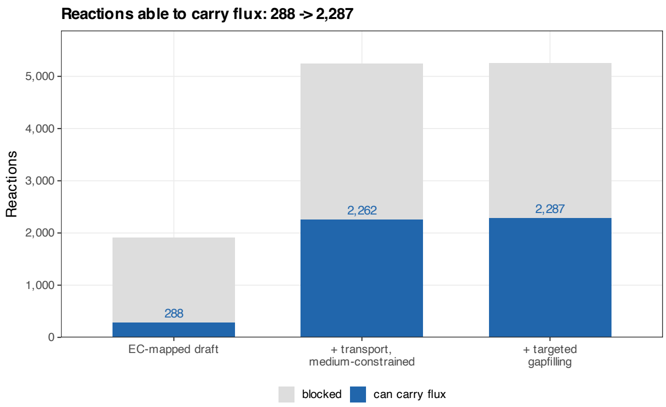

Fig. S5 Metabolic reconstruction by stage. Reactions able to carry flux rise from 288 in the EC-mapped draft to 2,287 after adding transport reactions, constraining exchanges to the culture medium, and targeted cofactor gapfilling.

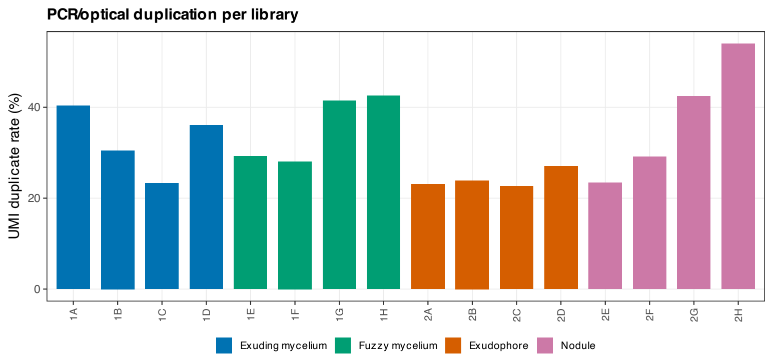

Fig. S6 UMI duplication rate per library.

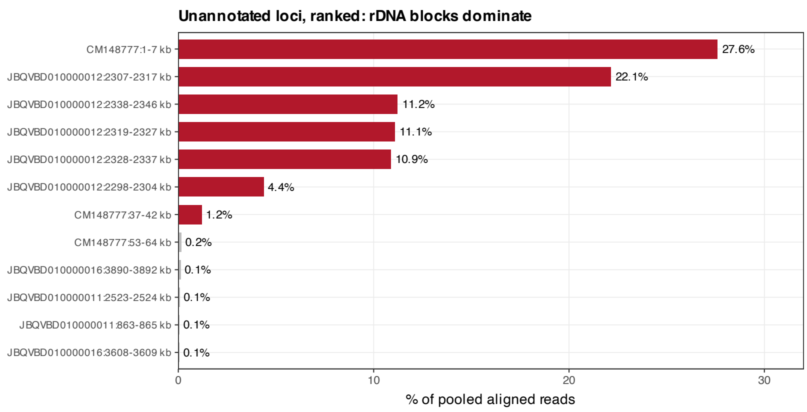

Fig. S7 Unannotated loci ranked by read share. Clustering reads that fall outside annotated genes; the ribosomal DNA blocks (red) dominate, which is how they were located.

Supplementary Tables

Table 4 Sample manifest. All libraries were sequenced on one flowcell and lane.

Library	Well	Tissue
GPNJ7M_1_sample_1A	1A	Exuding mycelium
GPNJ7M_2_sample_1B	1B	Exuding mycelium
GPNJ7M_3_sample_1C	1C	Exuding mycelium
GPNJ7M_4_sample_1D	1D	Exuding mycelium
GPNJ7M_5_sample_1E	1E	Fuzzy mycelium
GPNJ7M_6_sample_1F	1F	Fuzzy mycelium
GPNJ7M_7_sample_1G	1G	Fuzzy mycelium
GPNJ7M_8_sample_1H	1H	Fuzzy mycelium
GPNJ7M_9_sample_2A	2A	Exudophore
GPNJ7M_10_sample_2B	2B	Exudophore
GPNJ7M_11_sample_2C	2C	Exudophore
GPNJ7M_12_sample_2D	2D	Exudophore
GPNJ7M_13_sample_2E	2E	Nodule
GPNJ7M_14_sample_2F	2F	Nodule
GPNJ7M_15_sample_2G	2G	Nodule
GPNJ7M_16_sample_2H	2H	Nodule

Table 5 Differential expression across all six pairwise contrasts (12 retained libraries, rRNA-removed track; $p_{\mathrm{adj}} < 0.05$, $|\log_2 \mathrm{FC}| > 1$).

Tissue A	Tissue B	DE genes
Nodule	Fuzzy mycelium	239
Nodule	Exudophore	198
Nodule	Exuding mycelium	74
Fuzzy mycelium	Exudophore	50
Exudophore	Exuding mycelium	36
Fuzzy mycelium	Exuding mycelium	2

Table 6 Genome-scale metabolic reconstruction for *P. ostreatus* BOM_ss5.

Stage	Reactions	Metabolites	Genes	Carry flux	Gapfilled
EC-mapped draft	1,915	2,028	881	288	0
Transport + MNM medium	5,243	5,140	881	2,262	0
Targeted gapfilling	5,247	5,140	881	2,287	3

Gapfilled reactions carry no gene association.

Supplementary data files (comma/tab separated, in `latex/tables/`): S3 replicate-supported markers; S4 co-expression module titles; S5 module enrichment; S6 CAZyme assignments; S8 predicted secretome.

Data availability. Code, count matrices, annotation, models and all intermediate results: https://github.com/dr-richard-barker/Myco_tissue_RNAseq. Raw reads: SRA accession pending (submission metadata prepared in `submission/`). Archived release: Zenodo DOI pending.

Author contributions. [AUTHORS: CRediT statement for all eight authors.]

Funding. [AUTHORS: grant numbers; must match the BioProject record.]

Competing interests. [AUTHORS: this manuscript reports quality problems in a commercially supplied dataset; state the commercial relationship or its absence.]

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